

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Systems for Data Science, Performance Engineering, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

TypeScript, JavaScript, Python, SQL, Java, C++, HTML/CSS, Bash

Frontend

React, Redux, Next.js, responsive UI, REST APIs, web GUIs

Backend & Data

Python (FastAPI), Node.js, Express, REST APIs, event-driven / async services, PostgreSQL, DynamoDB, MongoDB, Redis

AI-Assisted & Agentic Dev

Cursor, Claude / Claude Code, GitHub Copilot, ChatGPT, LLM agents, MCP, prompt engineering, validating AI-generated output

Cloud & DevOps

AWS (S3, KMS, SQS), GCP, Azure, Docker, Git, GitHub Actions, CI/CD, Linux

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Built a full-stack web app end-to-end at a **healthcare** company, a data annotation platform (**React** frontend, **Node.js** backend, MongoDB) used daily by the AI team to label data for production clinical models; replaced spreadsheet workflows with secure role-based REST APIs, cutting **100+ hours/month** of manual work and improving throughput **60%**.
- Engineered the **multi-tenant NHS eTriage platform** for **8+ UK clients** (Node.js, MongoDB) processing **10,000+ monthly submissions**, integrating with Mayden's iaptus platform via REST APIs and aggregation pipelines, contributing to **\$340K+** in revenue.
- Owned backend reliability across the stack: monitored production logs during releases and client integrations, shipped fixes for tenant-specific incidents, hardened authentication and rate limiting, and remediated **20+ VAPT findings** on sensitive clinical data.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js and Fastify, backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core codebase from **JavaScript to TypeScript** across multiple repositories and prototyped LLM-powered context-aware replies in a chat product, improving type safety, maintainability, and end-user response quality.

Projects

AgenticSearch - Agentic Pipeline with AI-Output Validation

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, OpenAI / Groq, SQLite, Docker

- Developed a multi-stage **AI agent** (Python) that plans typed retrieval, reranks pages, extracts entities with deterministic parsers before LLM fallback, **verifies AI-generated outputs**, and returns cell-level provenance for every result.
- Hardened the service with **150+ automated tests**, provider routing/fallback across LLM backends, prompt iteration, and pipeline instrumentation, validating and debugging AI output before it reaches the user.

cf-ai-research-scout - Event-Driven AI Agent

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers, Durable Objects, WorkflowEntrypoint, D1, WebSockets

- Built an AI agent on an **event-driven** architecture: it fetches web sources, runs multi-step research, and streams responses to the client over **WebSocket** in real time, deployed end-to-end on serverless infrastructure.
- Designed durable workflow boundaries so multi-step tasks survive process restarts and tool failures, with agent state and conversation history persisted server-side without an external database.

Underdogs Fitness - Production Full-Stack Web App

underdogsfitness.in

React + Redux, Node.js, Express, MongoDB, Stripe | solo, live in production 3+ years

- Built and deployed a **production full-stack web app** solo end-to-end (**React + Redux** frontend, Node.js/Express backend): auth, membership/attendance management, Stripe payments with a partial-payment ledger, and cron-based automation, live and in use for **3+ years**.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.